

VENKATA SAI ANNAM

Data Engineer III | Chennai, India

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SUMMARY

Data Engineer with 3+ years of experience designing and optimizing large-scale data pipelines and cloud-based data platforms. Strong expertise in Python, SQL, Apache Airflow, Databricks, Snowflake, Hive, and AWS, with hands-on experience in ETL and workflow orchestration. Proven track record of optimizing data processing, reducing pipeline runtime, improving SLA performance, and achieving significant cloud cost savings. Experienced in Snowpark migration, CI/CD automation, scalable pipeline design, and production support for high-volume enterprise data workloads.

TECHNICAL SKILLS

Languages : Scala, Java, Python, SQL, Pyspark

Data Engineering : Apache Spark, Databricks, Spark SQL, Apache Airflow, PySpark, Snowpark, ETL, Data Warehousing

Cloud : Microsoft Azure, Azure Databricks, ADLS Gen 2, Azure Key Vault

Database : Snowflake, MySQL, Delta Lake

Devops & Version Control : Git, Git Actions, CI/CD Pipeline

Concepts : Distributed Computing, Batch Processing, Performance Optimization, Workflow Orchestration

PROFESSIONAL EXPERIENCE :

Data Engineer III

Jul 2023- Present

NielsenIQ (NIQ) Chennai, India

- Designed and Implemented **PreExtract Component** that filters the BDL and RMS source data from approximately 30TB to 3TB, reducing the data volume by nearly 90% and decreasing downstream processing **time by 30%**.
- Optimized cloud resource utilization by processing only the required data resulting in approximately **\$48000** in annual cloud cost savings, representing around **10% reduction** in overall cloud cost.
- Improved pipeline stability by reducing job failures caused by midweek source table updates, reducing SLA completion time by **2 hours**, and ensuring timely delivery of BD outputs to clients.
- Optimized BD pipeline scheduling by segmenting **60 clients** and **164 datasets** in US region according to their D+9 and D+10 SLAs, automating processing across Tranche 1 and Tranche 2 with 100 distinct batches.
- Configured large datasets in Tranche 1 for weekend execution, leveraging readily available **Spot Instances** to reduce cloud costs, while scheduling smaller datasets on weekdays to ensure efficient resource utilization and on-time Delivery.
- Developed the Scope Builder component for US-region jobs to filter the **142TB** fact 260 Hive table down to less than 2TB of BD-enabled data, reducing processing time for large datasets by 6 hours across 100 batches.
- Optimized US-region data processing and cloud resource utilization, saving approximately **\$31k annually** in cloud costs.
- Automated Batch Finder validation to compare BD parallel processing datasets with the database refresh table, identify new or missing datasets and send post-merge email notifications, enabling timely delivery while reducing manual effort for the support team.

BD Next Gen Snowpark Migration :

- Collaborated with the Data Science team to optimize decomposition processing by migrating from **Databricks to Snowpark** and shifting from store-level to market-level coefficients calculations, significantly reducing processing complexity and compute requirements.
- Achieved a **50% reduction** in cloud processing costs and improved the end-to-end processing **SLA to 6 hours**
- Redesigned the Helper Component to process all datasets at the processing group level, **consolidating 100 parallel US runs** into a single aggregated Model and enabling separate PG-level runs for EU and CALATAM.
- Reduced cloud infrastructure costs by approximately **\$36000 per annum** and decreased reliance on Spot Instances during weekend BAU runs by eliminating unnecessary dataset-level processing dependencies.

- Designed and optimized scalable Apache Airflow DAGs for BD processing by implementing parallel decomposition workflows across Distribution, Price, Competition and Promotion with Sequential orchestration of Pre-Decomposition, Trend, Dollar and Validation stages reducing overall pipeline **runtime by 30%**.
- Executed tag-based versioned deployment to the snowflake VM through **Github Actions**, publishing the Premodel package to version specific directories via SSH.

Data Engineering Intern

Feb 2023 - May

2023

NielsenIQ (NIQ) Chennai, India - [Certificate](#)

- Resolved production issues and applied targeted enhancements during my internship, improving application reliability and user experience
- Successfully migrated the deployment pipeline from **Jenkins to Github Actions**, Automating CI/CD workflows and improving deployment efficiency, maintainability and reliability.
- Managed weekly BAU runs and improved Databricks cost efficiency by migrating small batch workloads from **E-series** to more cost effective **D-series** nodes optimizing resource utilization and reducing operational cloud costs.

PROJECTS :

Malicious URL detection

- Developed a machine learning model to identify malicious URLs using URL-based features, with Random Forest achieving better detection performance.

Sliding Window Protocol – Reliable Data Transmission

- Implemented the **Sliding Window Protocol** for reliable, ordered data transmission, managing packet flow and acknowledgments using fixed-size sequence numbers and sender/receiver buffers.

Certifications:

SnowPro Associate: Platform — [Snowflake](#)

Jan 2026

EDUCATION :

Sastra Deemed University, Thanjavur, Tamil Nadu

Aug 2019 - Aug 2023

B.Tech - Computer Science Engineering(GPA 8.58/10)

- Led peer-learning sessions on Operating Systems and Data Structures, mentoring junior students in supporting their technical interview preparation